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RESEARCH ARTICLE

ESAM predicts cardiovascular mortality in diabetic hemodialysis patients

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Abstract

Aim: To assess endothelial cell selective adhesion molecule (ESAM) as predictor of cardiovascular mortality in diabetic dialysis patients (DDPs).

Methods: ESAM, clinical and laboratory parameters were assessed in 73 DDP. Cardiovascular mortality was recorded in a 2 years' prospective observational study.

Results: Baseline ESAM was 17.1 (10.05–24.8) ng/ml and was correlated to phosphate ($r = -0.42$, $p = 0.008$), parathormone ($r = -0.36$, $p = 0.048$), albumin ($r = -0.24$, $p = 0.048$). ESAM significantly predicted cardiovascular death in univariate [HR = 1.03, 95% CI (1.006–1.054), $p = 0.01$] and multivariate [HR = 1.034, 95% CI (1.003–1.066), $p = 0.03$] Cox analysis. Time to cardiovascular death was shorter for patients with ESAM >12.44 ng/ml, $p = 0.0045$.

Conclusion: ESAM is an independent predictor of cardiovascular mortality in DDP.

Keywords

Adhesion molecule, atherosclerosis, endothelium, mineral metabolism

History

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Introduction

Chronic kidney disease is a major health issue with its global prevalence of >10% and extremely high morbidity and mortality, despite progress of renal replacement therapies and evidence-based management of hemodialysis patients. People with end-stage renal disease (ESRD) have a 10- to 20-fold increase in risk of death compared to age and sex-matched subjects, risk escalation being the most obvious in young people (Jha et al., 2013). The main cause of death in patients with ESRD is cardiovascular disease, accounting for roughly 50% of deaths, followed by infection and malignancy (de Jager et al., 2009). Type 2 diabetes is the leading etiology of ESRD in developed countries and contributes to the cardiovascular risk of ESRD patients with traditional risk factors such as dyslipidemia, obesity and smoking (Zoccali et al., 2003). However, as controlling for these factors failed to correct dismal outcome of these patients, the study of other processes which would explain their negative prognosis is a subject of ongoing research.

As such, endothelial dysfunction, which plays a major role in type 2 diabetes (van den Oever et al., 2010), is a significant

contributor to morbidity and mortality in hemodialysis patients (Zoccali et al., 2003). Endothelial dysfunction is linked to increased oxidative stress and inflammation, and to initiation and progress of atherosclerosis (Schalkwijk & Stehouwer, 2005).

Endothelial cell selective adhesion molecule (ESAM) is an important factor in assuring adequate function of endothelial cells (Hirata et al., 2001). ESAM is involved in mechanical stability of the endothelial layer (Hirata et al., 2001; Nasdala et al., 2002), normal (Ishida et al., 2003) and tumor (Cangara et al., 2010; Clasper et al., 2008) angiogenesis, thrombus formation (Stalker et al., 2009), permeability of the endothelial layer (Hara et al., 2009) and leukocyte diapedesis (Inoue et al., 2010). ESAM is found in serum as a soluble fraction (Rohatgi et al., 2009) and is increased in diabetic patients and in patients with reduced glomerular filtration rate (Kacso & Kacso, 2012). Soluble ESAM was already proven an important predictor of cardiovascular disease in the general population (Rohatgi et al., 2009), but not in other study populations. The aim of our study was to assess, for the first time to our knowledge, the relationship between soluble ESAM and cardiovascular mortality in a cohort of type 2 diabetes patients on hemodialysis.

Material and methods

Patients

We conducted a prospective observational study for which we screened all type 2 diabetes patients on chronic hemodialysis

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from three dialysis centers from Transylvania. Inclusion criteria were diagnosis of type 2 diabetes and treatment with hemodialysis for at least 6 months. Exclusion criteria were diagnosis of a definite non-diabetic kidney disease, diagnosis of CKD previously to diagnosis of diabetes, presence of acute illness, recent infection and neoplasia.

Laboratory analysis

Evaluation at baseline comprised clinical data (weight, height, blood pressure) and medication records as well as laboratory assessment. Blood was drawn from the arterial needle before start of dialysis with and without anticoagulant; a sample was also drawn immediately post-dialysis. Serum was separated by centrifugation 10 000 rotations/min for 3 min.

Samples for urea, creatinine, glycaemia, total- and high-density lipoprotein (HDL) cholesterol, triglycerides, uric acid, calcium, phosphate, blood iron (automated colorimetric enzymatic method), parathormone (PTH), ferritin (electrochemiluminescence immunoassay ECLIA method), standard hematology panel (automated flow-cytometry method) and C-reactive protein (CRP), albumin, glycated hemoglobin, transferrin (immunoturbidimetric method) were shipped and assessed right away by an authorized laboratory. One serum sample was frozen at -80°C . ESAM was determined in this sample using an ELISA assay (Uscn Life Science Inc., Wuhan, PR China) with minimum detection limit 0.118 ng/ml, intra-assay CV $<10\%$ and inter-assay variability $<12\%$. Blood iron and ferritin were used to calculate transferrin saturation. Low-density lipoprotein cholesterol (LDL) was calculated with the Friedwald formula (Friedewald et al., 1972). All the patients were dialyzed using high-flux dialysis membranes with bicarbonate-buffered dialysate 3 times a week. Dialysis efficiency was estimated using Daugirdas equation for assessing single-pool Kt/V (spKt/V). Dry weight was established on a trial-and-error basis. Patients were prospectively followed up for 24 months. Treatment changes were made by the attending physician according to guidelines and local policy. All-cause and cardiovascular mortality was recorded. Cardiovascular mortality was defined as heart failure, pulmonary edema, ischemic heart disease, arrhythmia and cerebrovascular disease. The causes of death were ascertained by review of the patients; medical records. The present study respects the principles of the Declaration of Helsinki and was approved by the local ethics committee at our institution. All the participants provided informed consent.

Statistical analysis

Statistical analysis was performed using SPSS 15.0 (Chicago, IL) and Microsoft EXCEL (Redmond, WA) applications. Normal distribution was tested with the Kolmogorov–Smirnov test, if sample size was >30 and with Shapiro–Wilk test, if not. Normally distributed variables are presented as mean $\pm$ SD and non-normally distributed ones as median (interquartile range), frequencies are presented as number (percentage). For comparison of two means, the Student *t*-test or Mann–Whitney test was used. For comparison of qualitative variables, the chi-squared test was used. For identifying correlations between two continuous variables, in case of outliers values or non-

linear relation, Spearman correlation coefficient (ρ) was used instead Pearson's correlation coefficient (r), which was used otherwise. This was followed by multivariate linear regression using the enter method.

Cox's proportional hazards regression analysis was used to examine the associations of baseline variables with cardiac events. Variables that had an independent influence on survival or were judged to potentially be confounding factors were included in a multivariate model. Hazard ratios and their 95% confidence intervals were calculated. When the parameter was found significant by Cox's proportional hazard model, Kaplan–Meier analysis was applied to compare two groups stratified by a cut-off. The cut-off was taken as the value which maximize the Youden index using receiver operating characteristics (ROC) curve analysis. $p \leq 0.05$ was considered statistically significant.

Results

Seventy-three patients, 48 (65.75%) men, mean age 60.80 ± 9.60 years were included in the study. Thirty-six (49.31%) patients had previous records of diabetic retinopathy and 39 (53.42%) patients of diabetic neuropathy. Characteristics of the patients are listed in Table 1.

At baseline, ESAM correlated inversely to serum phosphate ($r = -0.42$, $p = 0.008$), PTH ($r = -0.36$, $p = 0.048$), albumin ($r = -0.24$, $p = 0.048$). In multivariate regression, with ESAM as dependent variable, including parameters that were significant in univariate regression, none of these parameters significantly predicted ESAM. Results of multivariate regression are presented in Table 2.

Table 1. Characteristics of patients.

Parameter (UM)	Value
BMI (kg/m^2)	28.48 (25.88–32.22)
SBP (mmHg)	132.50 (120.00–150.00)
DBP (mmHg)	80.00 (70.00–80.00)
Glycaemia (mg/dl)	152.00 (116.75–185.50)
Diabetes duration (years)	13.00 (8.50–20.50)
Dialysis vintage (months)	21.00 (12–36)
Hb A1C (%)	6.90 (6.20–7.60)
HDL cholesterol (mg/dl)	38.82 ± 16.36
LDL cholesterol (mg/dl)	103.52 ± 41.55
Triglycerides (mg/dl)	173.00 (112.00–285.00)
Uric acid (mg/dl)	6.14 ± 1.18
Albumin (g/dl)	3.89 (3.60–4.10)
Calcium (mg/dl)	8.63 (8.27–9.10)
Phosphate (mg/dl)	4.86 (4.23–5.67)
PTH (ng/ml)	220.10 (97.10–380.95)
Hb (g/dl)	11.50 ± 1.67
Transferrin saturation (%)	28.65 (26.20–34.19)
Ferritin (ng/ml)	440.00 (335.00–555.63)
CRP (mg/dl)	0.64 (0.60–2.16)
Creatinine (mg/dl)	8.16 ± 2.34
Urea (mg/dl)	131.00 (108.00–165.25)
spKt/v	1.30 (1.22–1.41)
ESAM (ng/ml)	17.10 (10.05–24.8)

Normally distributed variables are presented as mean $\pm$ SD; non-normally distributed ones as median (interquartile range), frequencies as number (%).

BMI, body mass index; S, systolic; D, diastolic; BP, blood pressure; HbA1C, glycated hemoglobin; HDL, high-density lipoprotein; LDL, low-density lipoprotein; PTH, parathormone; Hb, hemoglobin; CRP, C reactive protein; sp, single pool; ESAM, endothelial cell selective adhesion molecule.

Sixteen patients died during follow-up and 5 were lost from study (transfer to another center or transplantation); and 12 patients died from cardiovascular causes. Baseline ESAM predicted cardiovascular mortality in univariate Cox regression analysis. Other significant predictors of fatal cardiovascular events were serum calcium and PTH. Results of univariate Cox regression are presented in Table 3.

For building the multivariate Cox model, we included parameters that were significant predictors of mortality in univariate analysis, but also parameters that influenced ESAM levels at baseline. The multivariate model for cardiovascular mortality is also presented in Table 3.

When assessing for all-cause mortality, ESAM was a significant predictor of the latter in univariate analysis [HR = 1.027, 95% CI (1.006–1.048), $p = 0.01$] but not in the multivariate Cox model. Time-to-event analysis of cardiovascular death was studied using the Kaplan–Meier method, stratifying the plasma ESAM concentration at a cut-off value derived from the ROC curve of 12.44 ng/ml. Patients with higher ESAM had a shorter survival than the other patients $p = 0.45$. Illustration of the Kaplan–Meier analysis is given in Figure 1.

Discussion

The main finding in our study is the fact that ESAM was an important predictor for cardiovascular mortality in crude analysis and in the multivariate model. Kaplan–Meier analysis showed that patients with higher ESAM according to the ROC-derived cut-off had shorter time to cardiovascular death than the other patients. ESAM also predicted all-cause mortality in univariate but not multivariate analysis. Another important finding in our study is the link of ESAM to parameters of mineral metabolism. To the best of our knowledge, this result has not been previously reported.

The negative predictive value of ESAM with regard to cardiovascular outcomes is in line with previous reports in the general population and is very likely related to endothelial dysfunction. Recruitment and migration of leukocytes across the vascular endothelium plays a crucial role in atherosclerotic plaque formation and progression and is dependent on endothelial function. ESAM might play a particular role in leukocyte diapedesis involving the para-cellular route. In murine models, ESAM has been shown to regulate early neutrophil transmigration at sites of vascular inflammation (Inoue et al., 2010; Wegmann et al., 2006), without affecting lymphocyte transmigration (Wegmann et al., 2006). Lymphocytes are known favor the trans-cellular route, mediated mainly by intercellular adhesion molecule. On the other hand, ESAM is selectively expressed at the level of tight

interendothelial cell junctions contributing to stability of the endothelial layer. Taken together, these facts suggest that ESAM, as a structural component of the tight junction has a role in controlling para-cellular transmigration of leukocytes (Petri & Bixel, 2006; Wegmann et al., 2006). Similarly, in inflamed (post-ischemic) liver, ESAM mediates neutrophil extravasation (Khandoga et al., 2009). ESAM also interferes with other functions of the endothelium that are relevant for atherosclerosis: it influences VEGF-mediated permeability of the endothelial layer (Hara et al., 2009), has a restraining role on thrombus growth and stability (Brass et al., 2008; Stalker et al., 2009) and is associated to oxidative stress (Kacso et al., 2014). For all these reasons, ESAM might play a role in atherosclerotic plaque development and progression. The first evidence that the circulating fraction of ESAM is associated to negative cardiovascular outcomes was provided by a cohort study in the general population – The Dallas Heart study (Rohatgi et al., 2009). In diabetic patients, circulating ESAM levels were associated with other negative outcomes related to complications of diabetes: progression of nephropathy (Park et al., 2014) and hemorrhagic complications after coronary stent placement (Kikkert et al., 2013). However, the predictive value of ESAM for mortality has not been studied in diabetic patients. We chose to explore this issue in the high-risk population of type 2 diabetes hemodialysis patients, assessing the predictive value of ESAM for cardiovascular mortality. We confirm the previously reported high values of ESAM in diabetic dialysis patients (Kacso & Kacso, 2012). These could be due to a decrease in ESAM excretion or catabolism with decreasing renal function, but also to an increased expression or shedding of the molecule, possible in relation to micro inflammatory environment and endothelial dysfunction.

Other parameters that predicted cardiovascular mortality were those of mineral metabolism: calcium and PTH. As mentioned earlier, ESAM was significantly correlated in univariate analysis to parameters of mineral metabolism. This might be only a fortuity association, the reflection of a high cardiovascular risk status, as disorders of the bone mineral metabolism are important determinants of vascular calcification and impact significantly on overall and cardiovascular mortality. Hypercalcemia has been repeatedly reported as independent predictor of mortality in hemodialysis patients (Block et al., 2004; Kalantar-Zadeh et al., 2006). Whereas severe hyperparathyroidism is generally associated to increased mortality, low PTH, expression of low bone turnover, is also associated with worse prognosis, probably mediated by progressing calcifications (Jean et al., 2011; Rhee et al., 2012). In hemodialysis patients, there is an important relationship between vascular calcifications and increased cardiac mortality (Moe & Chen, 2008). Mediocalcosis due to calcifications is a distinct pathologic process from atherosclerosis. Medial calcification reduces arterial elasticity, increases pulse wave velocity and favors left ventricular hypertrophy. However, vascular calcifications also favor atherosclerosis related events by decreasing coronary microcirculation, by favoring local inflammation and focal occlusion of arteries secondary to the plaque formation (Moe & Chen, 2008; Shioi et al., 2001). The difference between fatal cardiovascular outcomes due to vascular calcifications as opposed to those which are a consequence of atherosclerosis

Table 2. Multivariate regression with ESAM-dependent variable.

	Unstandardized coefficients	Sig.	95% confidence interval for B	
			Lower bound	Upper bound
Constant	53.34	0.03	6.38	100.30
Parathormone	−0.01	0.31	−0.03	0.01
Phosphate	0.90	0.69	−3.60	5.40
Albumin	−8.97	0.13	−20.57	2.63

Table 3. Results of univariate and multivariate Cox regression analysis.

Parameter	Univariate analysis		Multivariate model	
	<i>p</i>	HR (95% CI)	<i>p</i>	HR (95% CI)
Age	0.29	1.03 (0.97–1.10)		
Sex	0.47	1.63 (0.44–6.05)		
BMI	0.33	1.04 (0.96–1.14)		
SBP	0.99	1.00 (0.97–1.04)		
DBP	0.94	1.00 (0.93–1.09)		
Glycaemia	0.22	1.01 (0.99–1.01)		
HbA1C	0.10	0.65 (0.39–1.08)		
HDL cholesterol	0.39	0.98 (0.94–1.02)		
LDL cholesterol	0.84	0.99 (0.98–1.01)		
Triglycerides	0.60	0.99 (0.98–1.01)		
Uric acid	0.53	1.60 (0.99–1.01)		
Albumin	0.74	1.27 (0.30–5.24)	0.39	1.98 (0.41–9.56)
Calcium	0.03	4.62 (1.13–18.90)	0.39	1.92 (0.43–8.48)
Phosphate	0.49	0.73 (0.30–1.78)	0.77	1.13 (0.48–2.66)
PTH	0.01	0.99 (0.98–1.00)	0.05	0.99 (0.99–1.00)
Hb	0.84	1.04 (0.73–1.46)		
Transferrin saturation	0.79	0.98 (0.87–1.12)		
Ferritin	0.88	1.00 (0.99–1.00)		
CRP	0.40	1.88 (0.66–1.18)		
Creatinine	0.16	0.81 (0.61–1.08)		
Urea	0.78	1.00 (0.98–1.02)		
spKt/v	0.34	8.85 (0.95–826.62)		
ESAM	0.12	1.03 (1.01–1.05)	0.03	1.03 (1.00–1.07)
Dialysis duration	0.15	0.66 (0.38–1.16)		
Diabetes duration	0.88	0.99 (0.93–1.06)		

Odds of survival free of atherosclerosis related cardiovascular events are presented as hazard ratio (HR) and confidence intervals (95% CI) in univariate and multivariate analysis.
BMI, body mass index; S, systolic; D, diastolic; BP, blood pressure; HbA1C, glycated hemoglobin; HDL, high-density lipoprotein; LDL, low-density lipoprotein; PTH, parathormone; Hb, hemoglobin; CRP, C reactive protein; sp, single pool; ESAM, endothelial cell selective adhesion molecule.

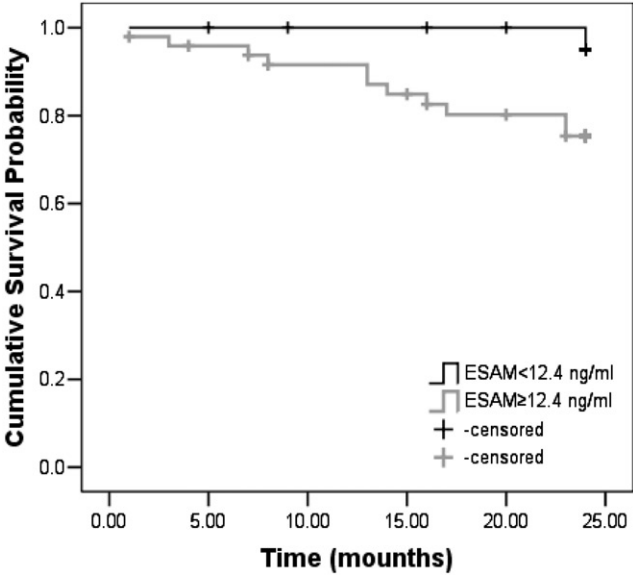


Figure 1. Kaplan–Meier survival curves for time to cardiovascular death according to baseline ESAM.

was not made in our study. ESAM and parameters of mineral metabolism might independently predict cardiovascular mortality, without a pathogenic interrelation. Nevertheless, direct implication of ESAM in calcification processes might be a complementary mechanism, not related to atherosclerosis, and warrants further study, as little is known of the intracellular cascade triggered by ESAM. Noteworthy, in the study of Rohatgi et al. (2009), the strongest association of

baseline ESAM was to aortic and coronary calcification and to measures of vascular stiffness, and a lesser association to aortic plaque burden.

Increased ESAM was associated in the general population to multiple risk factors for atherosclerosis including homeostasis model assessment index, presence of diabetes, increased BMI, blood pressure, inflammation, decreased renal function and to prevalent atherosclerotic complications (Rohatgi et al., 2009). This might be however, in the general population, only the expression of association of higher ESAM in diabetes (Kacso & Kacso, 2012) to the atherosclerosis risk factors of these patients. In our homogenous high-risk population, we could not identify such an association, except for the inverse relationship of ESAM to albumin that can be interpreted regarding albumin as a negative marker for inflammation. The fact that glycated hemoglobin is a suboptimal measure of glycemic control in dialysis patients might also be a factor explaining lack of correlation of ESAM with glycemic control (Rhee et al., 2014).

Conclusions

Endothelial cell selective adhesion molecule is an independent predictor for cardiovascular mortality in T2D patients. A link of ESAM to parameters of mineral metabolism is emerging.

Declaration of interest

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